WP8 Solutions

119. $z = \sin(3x)\cos(3y)$; $\frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial y} = \sin(3x)\cdot 3(-\sin(3y))$$
$$= -3\sin(3x)\sin(3y)$$

121. $z = \ln(x^6+y^4)$; $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial x} = \frac{1}{x^6+y^4}(6x^5)$$
$$= \frac{6x^5}{x^6+y^4}$$
$$\frac{\partial z}{\partial y} = \frac{1}{x^6+y^4}(4y^3)$$
$$= \frac{4y^3}{x^6+y^4}$$

125. $z = \tan(2x-y)$; $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial x} = \sec^2(2x-y)\cdot 2$$
$$= 2\sec^2(2x-y)$$
$$\frac{\partial z}{\partial y} = \sec^2(2x-y)(-1)$$
$$= -\sec^2(2x-y)$$

149. $z = e^x \sin y$

$$\frac{\partial z}{\partial x} = e^x \sin y$$
$$\frac{\partial^2 z}{\partial z^2} = e^x \sin y$$
$$\frac{\partial z}{\partial y} = e^x \cos y$$
$$\frac{\partial^2 z}{\partial y^2} = -e^x \sin y$$

so $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}$
$$= e^x \sin y - e^x \sin y$$
$$= 0.$$

157. $f(x,y,t) = e^{-2t}\sin x \sin y$

$$\frac{\partial f}{\partial t} = -2e^{-2t}\sin x \sin y$$
$$\frac{\partial f}{\partial x} = e^{-2t}\cos x \sin y$$
$$\frac{\partial^2 f}{\partial x^2} = -e^{-2t}\sin x \sin y$$
$$\frac{\partial f}{\partial y} = e^{-2t}\sin x \cos y$$
$$\frac{\partial^2 f}{\partial y^2} = -e^{-2t}\sin x \sin y$$
$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = -e^{-2t}\sin x \sin y + (-e^{-2t}\sin x \sin y)$$
$$= -2e^{-2t}\sin x \sin y$$
$$= \partial f/\partial t \quad \checkmark$$

161. $f(x,y) = 200x^{0.7}y^{0.3}$; $x = 500, y = 1000$

$$\partial f/\partial x = 200(0.7)x^{-0.3}y^{0.3}$$
$$\left.\frac{\partial f}{\partial x}\right|_{(500,1000)} = \frac{140(1000)^{0.3}}{500^{0.3}}$$
$$= 140(2)^{0.3}$$
$$\frac{\partial f}{\partial y} = 200(0.3)x^{0.7}y^{-0.7}$$
$$\left.\frac{\partial f}{\partial y}\right|_{(500,1000)} = \frac{60(500)^{0.7}}{1000^{0.7}}$$
$$= 60/2^{0.7}$$

171. $z = -9x^2 - 3y^2$; $P(2,1,-39)$

$f_x = -18x$ $\quad$ $f_y = -6y$

$f_x(2,1) = -36$ $\quad$ $f_y(2,1) = -6$

tangent: $z = -39 - 36(x-2) - 6(y-1)$
$$z = -3 - 36x + 72 - 6y + 6$$
$$36x + 6y + z = 39$$

181. $z = x^2 - 2xy + y^2$; $P(1,2,1)$

$f_x = 2x - 2y$ $\quad f_x(1,2) = 2(1) - 2(2) = -2$

$f_y = -2x + 2y$ $\quad f_y(1,2) = -2(1) + 2(2) = 2$

tangent: $z = 1 - 2(x-1) + 2(y-2)$

$z = 1 - 2x + 2 + 2y - 4$

$2x - 2y + z = -1$

196. $V(r,h) = \pi r^2 h$

$V_r = 2\pi r h$

$V_h = \pi r^2$

$dV = 2\pi r h\,dr + \pi r^2 dh$

197. $dV = 2\pi r h\,dr + \pi r^2 dh$

$d = 8cm$, $h = 12cm$, width $= 0.04cm$

$dV = 2\pi(4)(12)(0.04) + \pi(4)^2(0.04)$

$= 3.84\pi + 0.64\pi$

$= 4.48\pi\ cm^3$

203. $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$; $R_1 = 7\Omega$, $R_2 = 6\Omega$ and $\frac{\partial}{\partial r} = 0.05\Omega$

$R = \frac{R_1 R_2}{R_1 + R_2}$

$\frac{\partial R}{\partial R_1} = \frac{R_2^2}{(R_1+R_2)^2}$ $\quad \frac{\partial R}{\partial R_2} = \frac{R_1^2}{(R_1+R_2)^2}$

$dR\Big|_{(7,6)} = \frac{6^2}{(7+6)^2}(0.05) + \frac{7^2}{(7+6)^2}(0.05)$

$= \frac{(36+49)(0.05)}{13^2}$

$= \frac{85(.05)}{169}$

$dR = 0.025$

207. $f(x,y) = x\sqrt{y}$ $\quad P(1,4)$

$f_x = \sqrt{y}$ $\quad f_x(1,4) = 2$

$f_y = \frac{x}{2\sqrt{y}}$ $\quad f_y(1,4) = 1/4$

$L(1,4) = 1\sqrt{4} + 2(x-1) + \frac{1}{4}(y-4)$

$= 2 + 2x - 2 + \frac{1}{4}y - 1$

$= 2x + \frac{1}{4}y - 1$